Цителадзе Г.А.

Использовались модели GigaChat 3.5 Ultra, Claude Haiku 5.5 (medium) и GPT-6.1 Sol (hight)

Статья была “Реализация метода главных компонент в вычислительной среде R”

https://cyberleninka.ru/article/n/realizatsiya-metoda-glavnyh-komponent-v-vychislitelnoy-srede-r.pdf

Для сравнения взяты название, аннотация и ключевые слова, а также 227 слов основного текста. В отправленном моделям файле авторского английского перевода не было

Глосарий

Русский термин	Перевод
метод главных компонент	principal component analysis
главная компонента	principal component
снижение размерности	dimensionality reduction
дисперсия	variance
среднеквадратичное отклонение	standard deviation
пакет	package
открытый исходный код	open source
графический интерфейс пользователя	graphical user interface

Для тестирования использовались следующие промты

Промпт 1: Переведи текст из прикреплённого PDF на английский язык. Выведи только перевод

Промпт 2: Переведи текст из прикреплённого PDF на английский язык. Используй указанные ниже переводы терминов последовательно во всём тексте. Сохраняй смысл, числа, обозначения и структуру. Выведи только перевод.
Глоссарий:

- метод главных компонент — principal component analysis
- главная компонента — principal component
- снижение размерности — dimensionality reduction
- дисперсия — variance
- среднеквадратичное отклонение — standard deviation
- пакет — package
- открытый исходный код — open source
- графический интерфейс пользователя — graphical user interface

Сравнение с авторским переводом

Модель	Промпт	BLEU %	Косинус	Термины
GigaChat	1	19.4	0.864	7.0 из 9
Claude Haiku	1	17.1	0.838	7.5 из 9
Gpt6.1Sol	1	23.9	0.857	8.5 из 9
GigaChat	2	30.4	0.928	7.5 из 9
Claude Haiku	2	27.9	0.914	8.5 из 9
Gpt6.1Sol	2	27.5	0.891	9.0 из 9

Критерий проверки терминов: 1 - общепринятый термин, 0,5 - смысл понятен, но выражение неточное, 0 - неверный перевод.

Русский термин	Запуск	Перевод модели	Оценка
метод главных компонент	GigaChat, П1	principal component method	0.5
метод главных компонент	Claude Haiku, П1	principal component method	0.5
метод главных компонент	Gpt6.1Sol, П1	principal component analysis	1
метод главных компонент	GigaChat, П2	principal component analysis	1
метод главных компонент	Claude Haiku, П2	principal component analysis	1
метод главных компонент	Gpt6.1Sol, П2	principal component analysis	1
главная компонента	GigaChat, П1	principal component	1
главная компонента	Claude Haiku, П1	principal component	1
главная компонента	Gpt6.1Sol, П1	principal component	1
главная компонента	GigaChat, П2	principal component	1
главная компонента	Claude Haiku, П2	principal component	1
главная компонента	Gpt6.1Sol, П2	principal component	1
снижение размерности	GigaChat, П1	dimensionality reduction	1
снижение размерности	Claude Haiku, П1	dimensionality reduction	1
снижение размерности	Gpt6.1Sol, П1	dimensionality reduction	1
снижение размерности	GigaChat, П2	dimensionality reduction	1
снижение размерности	Claude Haiku, П2	dimensionality reduction	1
снижение размерности	Gpt6.1Sol, П2	dimensionality reduction	1
пакет	GigaChat, П1	packages	1
пакет	Claude Haiku, П1	packages	1
пакет	Gpt6.1Sol, П1	packages	1
пакет	GigaChat, П2	packages	1
пакет	Claude Haiku, П2	packages	1
пакет	Gpt6.1Sol, П2	packages	1
открытый исходный код	GigaChat, П1	open-source	1
открытый исходный код	Claude Haiku, П1	open-source	1
открытый исходный код	Gpt6.1Sol, П1	open-source	1
открытый исходный код	GigaChat, П2	open source	1
открытый исходный код	Claude Haiku, П2	open source	1
открытый исходный код	Gpt6.1Sol, П2	open source	1
графический интерфейс пользователя	GigaChat, П1	graphical user interface	1

Русский термин	Запуск	Перевод модели	Оценка
графический интерфейс пользователя	Claude Haiku, П1	graphical user interface	1
графический интерфейс пользователя	Gpt6.1Sol, П1	graphical user interface	1
графический интерфейс пользователя	GigaChat, П2	graphical user interface	1
графический интерфейс пользователя	Claude Haiku, П2	graphical user interface	1
графический интерфейс пользователя	Gpt6.1Sol, П2	graphical user interface	1
рейтинговый функционал	GigaChat, П1	rating function	0.5
рейтинговый функционал	Claude Haiku, П1	rating functional	1
рейтинговый функционал	Gpt6.1Sol, П1	ranking score	0.5
рейтинговый функционал	GigaChat, П2	rating function	0.5
рейтинговый функционал	Claude Haiku, П2	rating functional	1
рейтинговый функционал	Gpt6.1Sol, П2	ranking functional	1
корреляционное поле	GigaChat, П1	graph of correlation field	0.5
корреляционное поле	Claude Haiku, П1	correlation field plot	0.5
корреляционное поле	Gpt6.1Sol, П1	scatter plot	1
корреляционное поле	GigaChat, П2	graph of correlation field	0.5
корреляционное поле	Claude Haiku, П2	correlation scatter plot	1
корреляционное поле	Gpt6.1Sol, П2	scatter plot	1
прямая отрицательная корреляционная зависимость	GigaChat, П1	direct negative correlation	0.5
прямая отрицательная корреляционная зависимость	Claude Haiku, П1	direct negative correlation	0.5
прямая отрицательная корреляционная зависимость	Gpt6.1Sol, П1	negative linear correlation	1
прямая отрицательная корреляционная зависимость	GigaChat, П2	direct negative correlation dependence	0.5
прямая отрицательная корреляционная зависимость	Claude Haiku, П2	direct negative correlation	0.5
прямая отрицательная корреляционная зависимость	Gpt6.1Sol, П2	negative linear correlation	1

Оценки 0,5: “principal component method” понятно, но общепринятое название метода – “principal component analysis”. “Rating function” и “ranking score” неточно передают “рейтинговый функционал”. “Correlation field” - буквальный перевод; для графика больше подходит “scatter plot”. Сочетание “direct negative correlation” двусмысленно: здесь имеется в виду отрицательная линейная корреляция. Оценок 0 не было.

Вывод

Без глоссария лучше всего справилась Gpt6.1Sol. Глоссарий помог моделям 1 и 2 исправить название метода. По проверенным терминам Gpt6.1Sol получила лучший результат в обоих запусках.

GigaChat, промпт 1

IMPLEMENTATION OF THE PRINCIPAL COMPONENT METHOD IN THE R COMPUTING ENVIRONMENT

Abstract. The paper considers the implementation of the principal component method in the R computing environment. University rating indicators were used as the initial data.

Keywords: R computing environment, analysis, visualization, multidimensional data, dimensionality reduction, principal component method.

The R environment is a software tool designed for the analysis of multidimensional data and their visualization, with powerful graphical capabilities. Among many other types of software, it is distinguished by the fact that it is an open-source environment developed within the framework of the free software development project (GNU), and the functionality of its base version can be significantly increased by applying an extended system of commands or packages, which are also freely available. This environment uses a command-line interface, but for ease of use, integrated development environments (IDEs) have also been created, such as RStudio, JGR, RKWard, SciViews-R, and R Commander. This paper considers the implementation of the principal component method in RStudio, which is written in C++ and uses the Qt framework for the graphical user interface.

Visualization of rating indicators is possible if the dimensionality of the initial feature space is reduced from nine indicators to two or three, then each university will be characterized by a two-dimensional or three-dimensional point in the Cartesian rectangular coordinate system. The principal component method was chosen as the dimensionality reduction method.

A high value of the correlation coefficient indicates that the rating function and the first principal component are strongly interrelated. The first principal component absorbs the maximum of the differences between universities. Thus, without knowing the method of calculating the rating function, the positions of universities in the rating table were restored.

On the graph of the correlation field between the first principal component and the value of the rating function, a strongly pronounced direct negative correlation is observed. It can also be noted that with an increase in the rating function, the value of the principal component decreases.

Claude Haiku, промпт 1

Here is the translation of the abstract and introduction:

Implementation of the Principal Component Method in the R Computing Environment

Abstract. This paper examines the implementation of the principal component method in the R computing environment. University rating indicators were used as the source data.

Keywords: R computing environment, analysis, visualization, multidimensional data, dimensionality reduction, principal component method.

R is a software tool designed for the analysis and visualization of multidimensional data, with powerful graphical capabilities. Among many other types of software, it stands out as an open-source environment developed within the framework of the GNU free software project. Its base functionality can be greatly extended through advanced command sets or packages, which are also freely available. This environment uses a command-line interface, but integrated development environments (IDEs) have been created for convenience, such as RStudio, JGR, RKWard, SciViews-R, and R Commander. This paper considers the implementation of the principal component method in RStudio, which is written in C++ and uses the Qt framework for its graphical user interface.

Visualizing the rating indicators is possible only if the dimensionality of the original feature space is reduced from nine indicators to two or three. Each university can then be represented as a two-dimensional or three-dimensional point in a Cartesian coordinate system. The principal component method was chosen as the dimensionality reduction technique. A high correlation coefficient indicates that the rating functional and the first principal component are strongly related. The first principal component captures the maximum amount of differences between universities. Thus, without knowing how the rating functional is calculated, the positions of the universities in the rating table were reconstructed.

The correlation field plot between the first principal component and the value of the rating functional shows a strongly expressed direct negative correlation. It can also be noted that as the rating functional increases, the value of the principal component decreases.

Gpt6.1Sol, промпт 1

IMPLEMENTATION OF PRINCIPAL COMPONENT ANALYSIS IN THE R COMPUTING ENVIRONMENT

Abstract. This paper examines the implementation of principal component analysis in the R computing environment. University ranking indicators were used as input data.

Keywords: R computing environment, analysis, visualization, multivariate data, dimensionality reduction, principal component analysis.

R is a software environment designed for the analysis and visualization of multivariate data, offering powerful graphics capabilities. Among the many other types of software, it stands out as an open-source environment developed as part of the GNU free software project. The functionality of its base version can be significantly extended through additional commands or packages, which are also freely available. R uses a command-line interface, but integrated development environments (IDEs), such as RStudio, JGR, RKWard, SciViews-R, and R Commander, have also been created to make it easier to use. This paper examines the implementation of principal component analysis in RStudio, which is written in C++ and uses the Qt framework for its graphical user interface.

The ranking indicators can be visualized by reducing the dimensionality of the original feature space from nine indicators to two or three. Each university can then be represented as a two-dimensional or three-dimensional point in a Cartesian coordinate system. Principal component analysis was selected as the dimensionality reduction method.

The high magnitude of the correlation coefficient indicates that the ranking score and the first principal component are strongly related. The first principal component captures the greatest variation among universities. Thus, the universities' positions in the ranking table were reconstructed without knowing how the ranking score was calculated.

The scatter plot of the first principal component against the ranking score shows a pronounced negative linear correlation. It can also be observed that the value of the principal component decreases as the ranking score increases.

GigaChat, промпт 2

IMPLEMENTATION OF PRINCIPAL COMPONENT ANALYSIS IN THE R COMPUTING ENVIRONMENT

Abstract. The paper considers the implementation of principal component analysis in the R computing environment. University rating indicators were used as the initial data.

Keywords: R computing environment, analysis, visualization, multidimensional data, dimensionality reduction, principal component analysis.

The R environment is a software tool designed for the analysis of multidimensional data and their visualization, with powerful graphical capabilities. Among many other types of software, it is distinguished by the fact that it is an open source environment developed within the framework of the free software development project (GNU), and the functionality of its base version can be significantly increased by applying an extended system of commands or packages, which are also freely available. This environment uses a command-line interface, but for ease of work, integrated development environments (IDEs) have also been created, such as RStudio, JGR, RKWard, SciViews-R, R Commander. This paper considers the implementation of principal component analysis in RStudio, which is written in C++ and uses the Qt framework for the graphical user interface.

Visualization of rating indicators is possible if the dimensionality of the initial feature space is reduced from nine indicators to two or three, then each university will be characterized by a two-dimensional or three-dimensional point in the Cartesian rectangular coordinate system. Principal component analysis was chosen as the dimensionality reduction method.

A high value of the correlation coefficient indicates that the rating function and the first principal component are strongly interrelated. The first principal component absorbs the maximum of the differences between

universities. Thus, without knowing the method of calculating the rating function, the positions of universities in the rating table were restored.

On the graph of the correlation field between the first principal component and the value of the rating function, a strongly pronounced direct negative correlation dependence is observed. It can also be noted that with an increase in the rating function, the value of the principal component decreases.

Claude Haiku, промпт 2

PRINCIPAL COMPONENT ANALYSIS IMPLEMENTATION IN THE R COMPUTING ENVIRONMENT

Abstract. The paper considers the implementation of principal component analysis in the R computing environment. University rating indicators were used as the source data.

Keywords: R computing environment, analysis, visualization, multidimensional data, dimensionality reduction, principal component analysis.

R is a software environment designed for the analysis and visualization of multidimensional data, with powerful graphical capabilities. Among many other types of software, it stands out as an open source environment developed within the GNU free software project. Its basic functionality can be greatly extended through an expanded system of commands or packages, which are also freely available. This environment uses a command-line interface, but integrated development environments (IDEs) have also been created for convenience, for example RStudio, JGR, RKWard, SciViews-R, and R Commander. This paper considers the implementation of principal component analysis in RStudio, which is written in C++ and uses the Qt framework for its graphical user interface.

Visualization of the rating indicators is possible if the dimensionality of the original feature space is reduced from nine indicators to two or three, so that each university is characterized by a two- or three-dimensional point in a Cartesian rectangular coordinate system. Principal component analysis was chosen as the dimensionality reduction method.

A high value of the correlation coefficient indicates that the rating functional and the first principal component are strongly related. The first principal component captures the maximum of the differences between universities. Thus, without knowing how the rating functional is calculated, the positions of the universities in the rating table were restored.

The correlation scatter plot between the first principal component and the value of the rating functional shows a strongly pronounced direct negative correlation. It can also be noted that as the value of the rating functional increases, the value of the principal component decreases.

Gpt6.1Sol, промпт 2

IMPLEMENTATION OF PRINCIPAL COMPONENT ANALYSIS IN THE R COMPUTING ENVIRONMENT

Abstract. This paper examines the implementation of principal component analysis in the R computing environment. University ranking indicators were used as the input data.

Keywords: R computing environment, analysis, visualization, multidimensional data, dimensionality reduction, principal component analysis.

R is a software environment designed for analyzing and visualizing multidimensional data, with powerful graphical capabilities. Among many other types of software, it stands out as an open source environment developed as part of the GNU free software project. The functionality of its base version can be significantly extended through an expanded set of commands or packages, which are also freely available. This environment uses a command-line interface, but integrated development environments (IDEs), such as RStudio, JGR, RKWard, SciViews-R, and R Commander, have also been created to make it easier to use. This paper examines the implementation of principal component analysis in RStudio, which is written in C++ and uses the Qt framework for its graphical user interface.

The ranking indicators can be visualized by reducing the dimensionality of the original feature space from nine indicators to two or three. Each university can then be represented by a two-dimensional or three-dimensional point in a Cartesian coordinate system. Principal component analysis was selected as the method for dimensionality reduction.

The high value of the correlation coefficient indicates that the ranking functional and the first principal component are strongly related. The first principal component captures the greatest differences between

universities. Thus, the universities' positions in the ranking table were reconstructed without knowing how the ranking functional was calculated.

The scatter plot of the first principal component against the value of the ranking functional shows a pronounced negative linear correlation. It can also be noted that as the ranking functional increases, the value of the principal component decreases.